Exam 1: Study Guide

This study guide lists skills you need to be able to do for your exam. It does not tell you *how* to do the skills. Skills may be tested individually, together, or not at all. It is up to you to determine what skills you need already know, and which skills you need more practice on. Create a "Summary" page for each chapter with the key information you need to know.

Chapter 1 – Basics of Measurement

1) Express very large and very small numbers in scientific notation
   a) Section 1.1.1
2) Know SI units for length, mass, temperature, time and amount of a substance.
   a) Section 1.1.2
3) Know SI Prefix Multipliers
   a) Section 1.1.2
   b) Flashcards on Canvas (Print double-sided, cut on lines)
   c) Know kilo-, centi-, milli-, micro- and nano-
4) Report measured quantities to the correct number of digits (significant figures).
   a) Section 1.2.1
5) Determine which digits of a number are significant.
   a) Section 1.2.1-1.2.2
6) Round a number to an indicated number of significant figures, or indicated decimal place.
   a) Section 1.2.2
7) Determine the correct number of significant figures there are in the result of mathematical operations (addition, subtraction, multiplication and division).
   a) Section 1.2.2
8) Perform calculations involving unit conversions, given appropriate conversion factors ("Dimensional Analysis").
   a) Section 1.3.1-1.3.3
   b) Note: Become familiar with the Periodic Table/Exam Reference Sheet provided on Canvas. It is similar to your Yellow Sheet.
9) Calculate density, or use density as a conversion factor to calculate mass or volume of an item, given appropriate measured values.
   a) Section 1.4
10) Convert between temperature scales (°C , °F , K)
   a) Formulas will be given to you (like on your Yellow Sheet)

Chapter 2 – Atomic Structure

1) Define the following terms: observation, law, hypothesis, and theory
2) Define isotope, mass number, atomic number, and atomic mass.
   a) Section 2.3 (Atomic Number, Mass Number)
   b) Section 2.6 (Isotope)
   c) Section 2.7 (Atomic Mass)

3) Describe or draw the structure of an atom based on the nuclear model of an atom.
   a) Section 2.2, 2.8
   b) Know which subatomic particles exist in the nucleus
   c) Know charges of subatomic particles
   d) Know that most of the atom is empty space, and most of the mass is in the nucleus
4) Locate elements on the periodic table given name of group and period, or atomic number.
   a) Section 2.4
5) Classify elements as metals, non-metals or metalloids based on location of element on periodic table
   a) Section 2.4
   b) Which are likely to gain or lose electrons? (Section 2.5)
6) Define ion, cation, and anion
   a) Section 2.5
7) Write nuclear symbols for elements (including mass number, atomic number, and charge)
   a) Section 2.6
8) Determine the number of protons, neutrons, and electrons for a given isotope or ion.
   a) Section 2.6
9) Calculate the atomic mass of an element given percent abundance and mass of each isotope.
   a) Section 2.7

Chapter 3 – Light, Electrons, and the Periodic Table

1) Rank regions of the electromagnetic spectrum by wavelength, frequency, and energy.
   a) Section 3.1.1
2) Calculate frequency or wavelength of light ($c = \lambda \nu$)
   a) Section 3.1
3) Describe key differences between the Bohr model and Quantum Mechanical Model of the atom.
   a) Section 3.2.1, 3.2.2
4) Describe how the Bohr model can be used to explain atomic spectra (emission spectra).
   a) Section 3.2.1
5) Identify the number of subshells, orbitals, and electrons in a given principle shell (n).
   a) Section 3.3
6) Identify which orbital box diagram follows electron filling rules:
   a) Section 3.3
   b) Aufbau: Fill lowest energy to highest energy
   c) Hund's Rule: For orbitals of the same energy, fill one electron per orbital before pairing.
   d) Pauli Exclusion Principle: Each orbital can hold a maximum of two electrons, with opposite spins
7) Identify the number of valence or core electrons in an atom or ion.
8) Determine the number of unpaired electrons in an atom or ion by filling out an orbital box diagram
   a) Section 3.3
9) Determine the electron configuration for an atom or ion.
   a) Section 3.3
   b) (Full electron configuration or noble gas configuration)
10) Apply periodic trends for atomic radius, ionization energy, electron affinity, and metallic character to rank elements by those properties.
   a) Section 3.5